Henry Hsu

Houston, TX 77030

+1-510-996-6837 | www.yarikama.com | yarikama | yarikama | hsuhengjui@gmail.com

SUMMARY

Software Engineering Intern at Google and former GenAI Team Lead, shipping production GenAI platforms and agentic developer infrastructure. Grew a B2B platform's client base by 120% and user base to 20K via Agentic AI Systems, Advanced RAG, and backend infrastructure, earning two major industry awards. Now pursuing an M.C.S. at Rice University.

WORK EXPERIENCE

Software Engineering Intern — **Jun. 2026 – Aug. 2026**
Google — *Taipei, Taiwan*

- **Integration Test Infrastructure –** Built **3** test infrastructures from a **zero** baseline (Java, Python), onboarding **400+** integration test suites across **200+** smart home device types, with **90%+** unit test coverage for the infrastructure itself
- **Agentic Failure Triage –** Developed an LLM agent that triages scheduled integration test runs and surfaces root-cause summaries directly on issue tickets, replacing manual log inspection and cutting failure narrow-down time by **20+ minutes** per issue
- **Local Device Reproduction –** Extended the agent to reproduce device failures locally against virtual devices instead of waiting on cloud runs, cutting manual reproduction time by **85%**; **shipped and adopted by the team**

Generative AI Team Lead — **Dec. 2024 – Aug. 2025**
MaiAgent Co., Ltd. – *Award-Winning B2B GenAI Startup* — *Taipei, Taiwan*

- **Building AI Agents –** Upgraded chatbots into AI agents by integrating memory systems, tool APIs, and MCP Client, resulting in **120%** partner growth (CTBC Bank, MSI, HPE, iGroup) and user expansion from **3K** to **20K** users while cutting LLM token usage by **67%+**
- **Contributing to Open Source –** Contributed **14** merged pull requests to the **LlamaIndex** open-source project (**15K+ stars**), fixing bugs and adding features in integrations with AWS Bedrock, Claude, Elasticsearch, Cohere, OpenAI, MCP Client, and AI Agent Workflow
- **Testing Framework –** Introduced pytest testing framework for unit and E2E coverage to prevent production regressions, achieving **67%** test coverage from **zero** baseline, which reduced production hotfixes by **90%** initially and sustaining a **50%** reduction long term
- **Technical Roadmap & Leadership –** Defined GenAI product roadmap for a 10-person startup; led feasibility evaluation and end-to-end delivery of production features including Agentic RAG, Artifact Generation, Tagging & Permission System, and Information Retrieval

Generative AI Intern — **Sep. 2024 – Dec. 2024**
MaiAgent Co., Ltd. – *Award-Winning B2B GenAI Startup* — *Taipei, Taiwan*

- **Pipeline Scaling –** Refactored with async framework and dual-database synchronization (RDB/Vector DB), achieving **3.5x** faster parsing and scaling from **3M** to **20M+** text chunks while **doubling** indexing performance
- **Reranker Integration –** Integrated Cohere and BGE open-source rerankers as a customer-selectable retrieval feature, improving RAG precision on large documents by **30% Precision@5** and **7% Precision@10**

EDUCATION

Rice University - Top 20 U.S. University — **Aug. 2025 – Dec. 2026 (Expected)**
M.C.S., Computer Science (GPA: 4.00/4.00) — *Houston, TX*

National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University — **Sep. 2020 – Jun. 2024**
B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List) — *Hsinchu, Taiwan*
Minor: Computer Science (4.13/4.30)

PROJECTS

Agentic Hybrid RAG | *Graph RAG (Neo4j, Cypher), Vector RAG (Milvus), Multi-Agent (LangGraph)* — **Feb. 2024 – Aug. 2024**

- **Self-Routing Agentic RAG:** Implemented a query classification agentic AI system that picks a global or local approach to request slicing from Graph RAG, Vector RAG or a Hybrid (Graph & Vector) RAG path to answer the contextual or detail-oriented queries
- **ML Clustering:** Applied ML clustering algorithms (K-means and DBSCAN) in recognizing pivotal nodes from vector database to build a low-cost Graph RAG; saved **35%** of indexing token cost, with only 2.56% worse in RAGAS metrics
- **Advanced Hybrid RAG:** Enhanced vector RAG hit rate using data retrieved from Graph RAG with HyDE, balancing both global and detailed information for more accurate responses. Achieved an MAR@10 of **88.2%** on multi-hop datasets

AWARDS

Presidential Hackathon Winner (2024) - Urban noise detection with LLM-powered structured analysis
23rd Golden Peak Award (2025) - Outstanding Commercial Product, MaiAgent AI Platform

SKILLS

Languages: Python, Java, C/C++, TypeScript | **GenAI:** LlamaIndex, LangChain, LangGraph, MCP | **Fullstack:** Django, FastAPI, React
Databases: PostgreSQL, MongoDB, Milvus, Elasticsearch | **Tools:** nvim, git, Docker, Shell Script, Nginx | **Methodologies:** Agile, Scrum
Data Science: numpy, pandas, scikit-learn, PyTorch, JAX | **AWS:** Bedrock, EMR, EC2, S3, RDS, ElastiCache, SageMaker